

Multiple Choice

1. **Answer:** A

Options: A) Levels of oxygen, acidity or alkalinity of the medium; B) Temperature changes, light exposure; C) Atmospheric pressure, noise levels; D) Nitrogen levels, radiation exposure; E) Soil texture, light wavelength

Note: The correct answer highlights that bacteria not only require specific temperature ranges for optimal growth but also depend on the availability of oxygen and the pH level of their environment, as these factors significantly influence their metabolic processes and overall viability.

2. **Answer:** C

Options: A) Lysosomes act as storage units for excess nutrients; B) Lysosomes are used for energy production; C) Lysosomes function in intra cellular digestion; D) Lysosomes synthesize proteins for the cell; E) Lysosomes regulate cell division

Note: Lysosomes contain digestive enzymes that break down waste materials and cellular debris, making them essential for intracellular digestion and maintaining cellular health.

3. **Answer:** A

Options: A) Only two DNA regions, V and D, combine in heavy chains; B) V, D, and JH regions are combined in light chains; C) V and JH regions are combined, while D regions are excluded from heavy chains; D) Five separate DNA regions are combined in heavy chains, adding extra complexity; E) D and JH regions are combined, while V regions are excluded from heavy chains

Note: The correct answer is based on the fact that in the heavy chains of antibodies, the variable region is generated by the recombination of only two specific DNA segments, known as the variable (V) and diversity (D) gene segments, which contributes to the vast diversity of antibody sequences necessary for effective immune responses.

4. **Answer:** A

Options: A) 12.5 percent; B) 40 percent; C) 18.75 percent; D) 25 percent; E) 37.5 percent

Note: The probability of getting exactly three heads in five flips of a balanced coin is calculated using the binomial probability formula, which yields a result of 12.5 percent when considering the number of favorable outcomes (10) over the total possible outcomes (32).

5. **Answer:** E

Options: A) -.09176; B) .21185; C) .37748; D) .55962; E) .65361

Note: The correct answer of .65361 is derived from calculating the area under the normal distribution curve between the values of 5 and 11, using the z-scores corresponding

to these values, which reflects the probability of a normally distributed variable falling within that range.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Directional selection can lead to distinct color patterns in species, such as certain African butterflies, because it favors individuals with specific color traits that enhance survival and reproduction in their particular environments, thereby gradually shifting the population's color distribution.

7. **Answer:** True

Options: A) True; B) False

Note: Natural selection primarily favors the inheritance of genes associated with advantageous traits that enhance survival and reproduction, while genes linked to less advantageous or 'unfit' traits tend to diminish in frequency over generations.

8. **Answer:** False

Options: A) True; B) False

Note: The statement is false because following behavior in lambs is primarily instinctive and driven by their innate need for safety and social bonding, rather than learned through observation and imitation of actions.

9. **Answer:** True

Options: A) True; B) False

Note: Clathrin is primarily known for its role in vesicular transport and endocytosis, while fibroblast motility relies on adhesion plaques, vinculin, and lamellipodia for cell movement, thus confirming that clathrin is not directly involved in this specific process.

10. **Answer:** True

Options: A) True; B) False

Note: The statement is true because if the enzymes responsible for breaking down acetylcholine, such as acetylcholinesterase, are denatured, they will be unable to hydrolyze acetylcholine in the synaptic cleft, leading to its accumulation.

Long Form Response

11. **Answer:** The significant variations in the size of the human appendix among individuals challenge the notion that it is a completely vestigial organ. If the appendix were entirely redundant, one would expect it to be uniformly small or absent across the population; however, its considerable size differences suggest that it may still serve a functional purpose. Research indicates that the appendix might play a role in gut flora maintenance and immune function, particularly in young individuals. This variability implies that the organ may have adapted to various dietary and environmental

conditions, pointing to a more complex evolutionary history than the vestigial model suggests. Consequently, the size of the appendix serves as evidence against the idea of its complete redundancy in human physiology.

Note: correct because the observed size variations of the human appendix among individuals imply that it may retain functional roles, such as supporting gut health and immune response, rather than being entirely vestigial.

12. **Answer:** A flock of birds exemplifies an elementary form of society known as a motion group, characterized by collective movement and social interaction. This classification highlights the cooperative behaviors observed in birds, such as flocking for foraging, predator avoidance, and navigation. By studying these motion groups, researchers gain insights into social dynamics, communication strategies, and the evolutionary advantages of group living in animal behavior. Additionally, understanding the organization and structure of such groups can inform broader ecological and evolutionary theories, illustrating how social behaviors contribute to survival and reproduction in various species.

Note: correct because it accurately describes a flock of birds as a motion group that demonstrates cooperative behaviors essential for survival, while also emphasizing the significance of studying these social dynamics to understand animal behavior and the evolutionary benefits of group living.